Tutorial 07

ENVX2001 – Applied Statistical Methods

Semester 1

Welcome to week 7!

This week's lecture introduced you to Simple- and Multiple- Linear Regression. For both of these regression techniques we defined the key steps in model fitting and how to interpret model output.

In this tutorial, we will explore key differences between simple linear regression (SLR) and multiple linear regression (MLR), and we will learn how to fit these models in R. We will also discuss how to test assumptions and evaluate model fit for both types of regression.

You will learn how to:

1. Distinguish key similarities and differences between simple linear regression (SLR) and multiple linear regression (MLR) models
2. Implement the fitting process for each type of model
3. Test assumptions and evaluate model fit of SLR and MLR models.

Simple linear regression vs Multiple Linear Regression

Table 1: Similarities and differences between MLR and SLR models

Element	Simple Linear Regression (SLR)	Multiple Linear Regression (MLR)
Number of predictors	1	2 or more
Model form	$Y = \beta_0 + \beta_1 x + \epsilon$	$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n + \epsilon$
How to fit in R	`lm(y ~ x, data = dataset)`	`lm(y ~ x1 + x2 + ... + xn, data = dataset)`
Assumptions	L.I.N.E.	L.I.N.E., collinearity
Model evaluation	r^2, residual plots	Adjusted r^2, residual plots, VIF
Conclusion	"as x increases, y also increases/decreases by β_1 units"	"as x_1 increases, y also increases/decreases by β_1 units, holding all other predictors constant"

Setting up for our analysis

Packages

This tutorial uses `tidyverse`, `readxl`, `moments`, `psych`, and `performance`. Install any you are missing by running the following **in the console**:

```
CODE
install.packages(c("tidyverse", "readxl", "moments", "psych", "performance"))
```

If a package is already installed, R will simply reinstall it — no harm done.

The data

Figure 1: Black Cherry tree foliage and flowers, from Wikimedia Commons, by Sten Porse.

For both examples we will use the *trees* dataset, which is an in-built dataset provided by R. It contains measurements of diameter (inches), height (feet) and volume of timber (cubic feet) of 31 felled Black Cherry trees.

`Volume` will be our response variable for both models.

Note: the 'girth' variable within the dataset is actually the diameter of the tree. We can rename it now, which also adds it into our R Environment:

CODE

```r
trees <- trees %>%
    rename(Diameter = Girth)

head(trees)
```

OUTPUT

```
  Diameter Height Volume
1      8.3     70   10.3
2      8.6     65   10.3
3      8.8     63   10.2
4     10.5     72   16.4
5     10.7     81   18.8
6     10.8     83   19.7
```

Original data source:

Meyer H. A. (1953). Forest Mensuration. Penns Valley Publishers.

Ryan T. A., Joiner B. L., Ryan B. F. (1976). The Minitab Student Handbook. Duxbury Press, North Scituate, MA. ISBN 087150359X

Analysis

Exercise 1: State the models and hypotheses

Part A — Simple linear regression

The statistical model for a simple linear regression is:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Where:

- y_i is the response variable (Volume) for the ith observation when the predictor X = x_i (Diameter)
- β_0 is the intercept where the regression line crosses the y-axis (the expected timber Volume when Diameter is 0)
- β_1 is the slope of the regression line, which represents the change in timber Volume for a one-unit increase in Diameter
- ϵ_i is the random error term for the ith observation and is assumed to be normally distributed with mean 0 and constant variance

OR if written in terms of the variables in our dataset, the model would be:

$$Volume = \beta_0 + \beta_1 \cdot Diameter + \epsilon_i$$

$H_0 : \beta_1 = 0$, The slope is equal to zero, i.e. there is no relationship between Diameter and Volume

$H_1 : \beta_1 \neq 0$, The slope is not equal to zero, i.e. There is a relationship between diameter and volume

Part B — Multiple linear regression

The statistical model for a multiple linear regression with 2 predictors is:

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon_i$$

Where:

- y_i is the response variable (timber volume) for the ith observation
- β_0 is the intercept where the regression line crosses the y-axis
- β_1 is the slope of the regression line for the first predictor, which represents the change in Volume for a one-unit increase in tree Diameter
- β_2 is the slope of the regression line for the second predictor, which represents the change in Volume for a one-unit increase in tree Height
- ϵ_i is the random error term for the ith observation and is assumed to be normally distributed with mean 0 and constant variance

OR if written in terms of the variables in our dataset, the model would be:

$$Volume = \beta_0 + \beta_1 \cdot Diameter + \beta_2 \cdot Height + \epsilon_i$$

For Multiple linear regression we are testing two (or more) predictors, so we have multiple null and alternative hypotheses to test.

First we test the partial regression coefficients for each predictor:

$H_0 : \beta_1 = 0$, The partial regression slope is equal to zero, i.e. there is no relationship between Diameter and Volume

$H_1 : \beta_1 \neq 0$, The partial regression slope is not equal to zero, i.e. There is a relationship between diameter and Volume

$H_0 : \beta_2 = 0$, The partial regression slope is equal to zero, i.e. there is no relationship between Height and Volume

$H_1 : \beta_2 \neq 0$, The partial regression slope is not equal to zero, i.e. There is a relationship between Height and Volume

Then we also test the overall model:

$H_0 : \beta_1 = \beta_2 = 0$ (all partial regression slopes are equal to zero, i.e. there is no relationship between diameter, height and volume)

$H_1: \beta_1 \neq 0$ and/or $\beta_2 \neq 0$ (at least one partial regression slope is not equal to zero, i.e. there is a relationship between either diameter and volume, or height and volume).

i Checkpoint

You should now be able to write down the model equation for both simple and multiple linear regression, and state the associated null and alternative hypotheses. You should also be able to explain why multiple linear regression requires a separate null hypothesis for each partial regression coefficient, as well as a null hypothesis for the overall model.

Exercise 2: Explore the data

Part A — Distributions

For each variable we need to check the distribution, identify outliers and check for correlations between predictors (for MLR).

CODE
```
library(moments)

summary(trees)
```

OUTPUT
```
    Diameter         Height       Volume
 Min.   : 8.30   Min.   :63   Min.   :10.20
 1st Qu.:11.05   1st Qu.:72   1st Qu.:19.40
 Median :12.90   Median :76   Median :24.20
 Mean   :13.25   Mean   :76   Mean   :30.17
 3rd Qu.:15.25   3rd Qu.:80   3rd Qu.:37.30
 Max.   :20.60   Max.   :87   Max.   :77.00
```

CODE
```
# Apply the sd function for each column
apply(trees, 2, sd) # (margin = 2 means apply the function to columns)
```

OUTPUT
```
 Diameter    Height    Volume
 3.138139  6.371813 16.437846
```

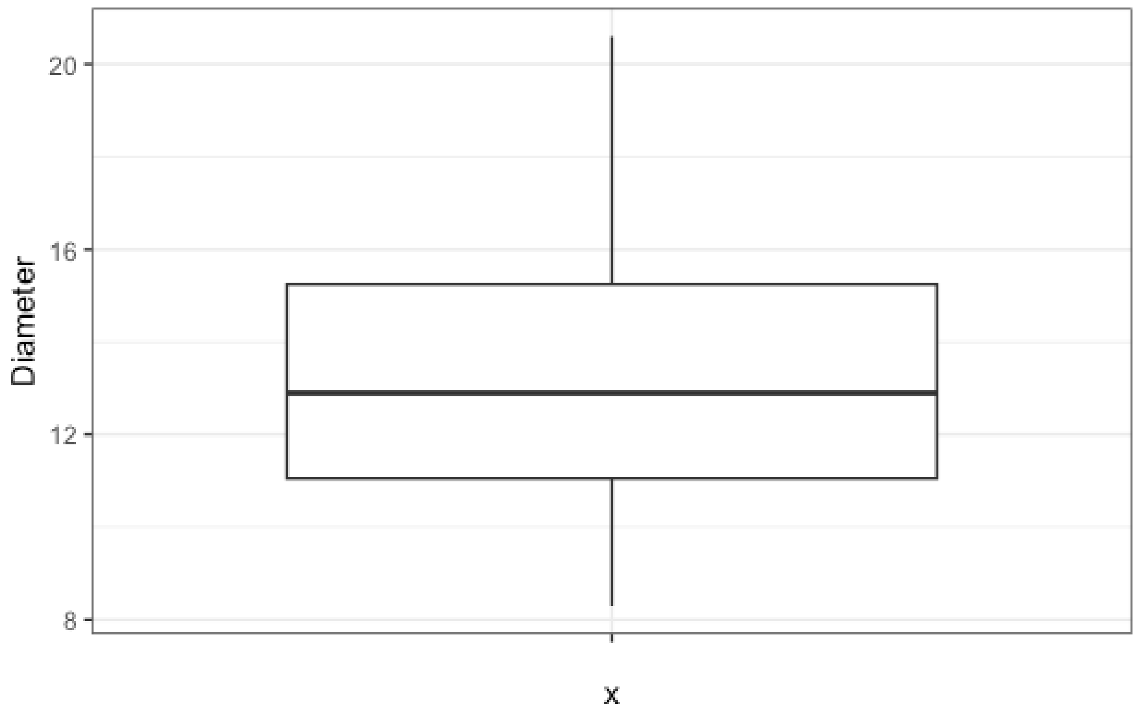
CODE
```
skewness(trees)
```

OUTPUT
```
  Diameter     Height     Volume
 0.5263163 -0.3748690  1.0643575
```

The mean diameter is 13.25 inches and the median is 12.90 inches. The values range from 8.30 to 20.60 inches and the standard deviation is 3.14 inches. There is a slight skew, indicated by the skewness coefficient of 0.526. The slight skew is also visible in the following boxplot where the median lies closer to the first quartile.

CODE

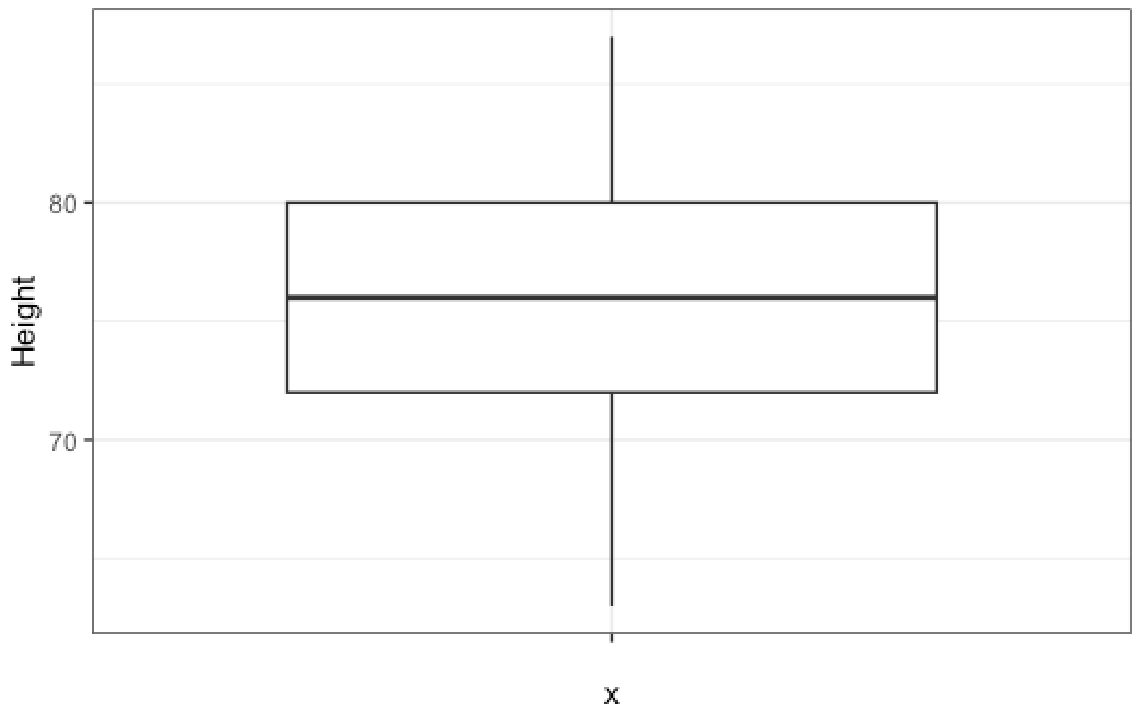
```
ggplot(trees, aes(x = "", y = Diameter)) +
    geom_boxplot() +
    theme_bw()
```



The mean and median height is 76 feet and the values range from 63 to 87 feet. The standard deviation is 6.37 feet and the skewness coefficient is −0.37, indicating a relatively symmetrical distribution. The boxplot also shows a relatively symmetrical distribution, with the median lying close to the centre of the box.

CODE

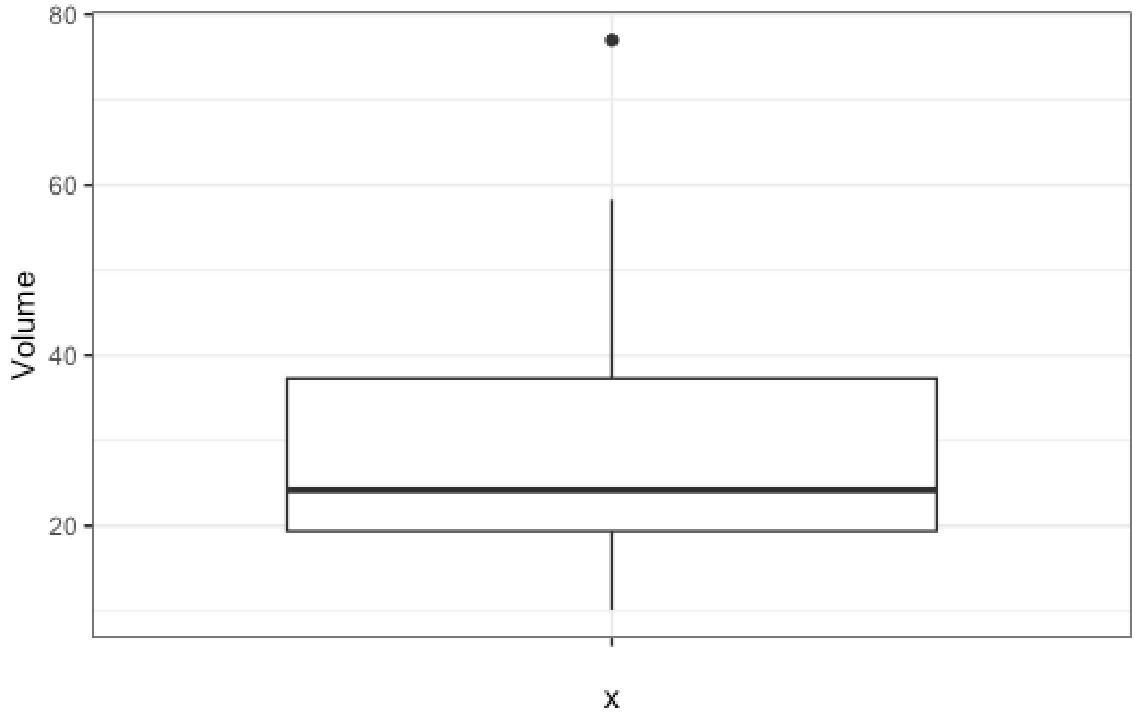
```
ggplot(trees, aes(x = "", y = Height)) +
    geom_boxplot() +
    theme_bw()
```



The mean volume of timber is 30.17 cubic feet and the median is 24.20 cubic feet. The values range from 10.20 to 77.00 cubic feet and the standard deviation is 16.44 cubic feet. The skewness coefficient is 1.06, indicating that the variable is positively skewed. This is also visible in the boxplot, where the median lies closer to the first quartile and there are some high values that are potential outliers.

CODE

```
ggplot(trees, aes(x = "", y = Volume)) +
    geom_boxplot() +
    theme_bw()
```



Why do the high values matter?

The high values would have high leverage in our model, causing the residuals to be skewed, violating our assumptions. The boxplot is therefore telling us that we might need to transform Volume to stabilise the variance of the regression, but it is worth confirming this with the diagnostic plots after fitting the model.

Part B — Correlations

For both simple and multiple linear regression we need to check the correlation between our response variable and our predictor variable.

For multiple linear regression it is also important to check the relationships between the predictor variables, as a strong correlation may indicate collinearity, which can influence the model.

We will start by looking at all correlations, and then for our simple linear regression we will select the variable that has the strongest correlation with the response (Volume) to use as our single predictor variable.

First we will use the correlation coefficient to understand the strength and direction of the relationships between the variables. Then we can look at the scatterplots between variables to verify the correlation coefficients and identify if there are any other types of relationships (e.g. non-linear) that may be occurring between variables.

Remember

Correlation coefficients only measure the strength and direction of LINEAR relationships.

It is therefore important to visualise the relationships between variables to check for non-linear relationships, which may not be captured by the correlation coefficient.

Let's have a look at the correlation coefficients between variables:

CODE

```r
cor(trees)
```

OUTPUT

```
          Diameter    Height    Volume
Diameter 1.0000000 0.5192801 0.9671194
Height   0.5192801 1.0000000 0.5982497
Volume   0.9671194 0.5982497 1.0000000
```

The correlation coefficient between Volume and Diameter is 0.967, which indicates a very strong positive linear relationship.

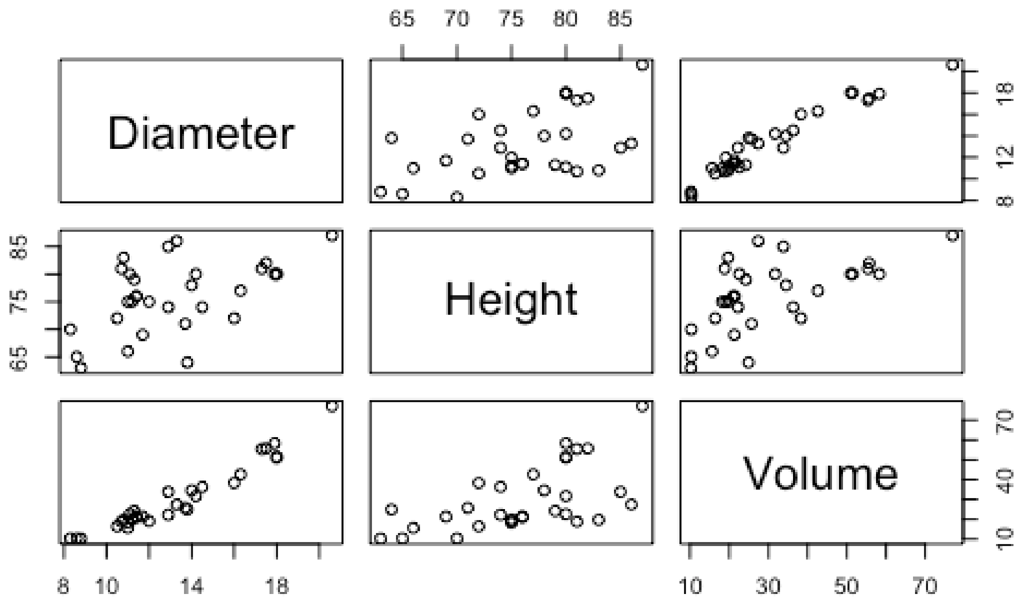
The correlation coefficient between Volume and Height is 0.598, which indicates a weak positive linear relationship.

The correlation coefficient between Diameter and Height is 0.519, which indicates a weak positive linear relationship, suggesting that there may be collinearity between these two predictor variables.

Now let's look at the scatterplots between variables to verify these relationships:

CODE

```r
pairs(trees)
```



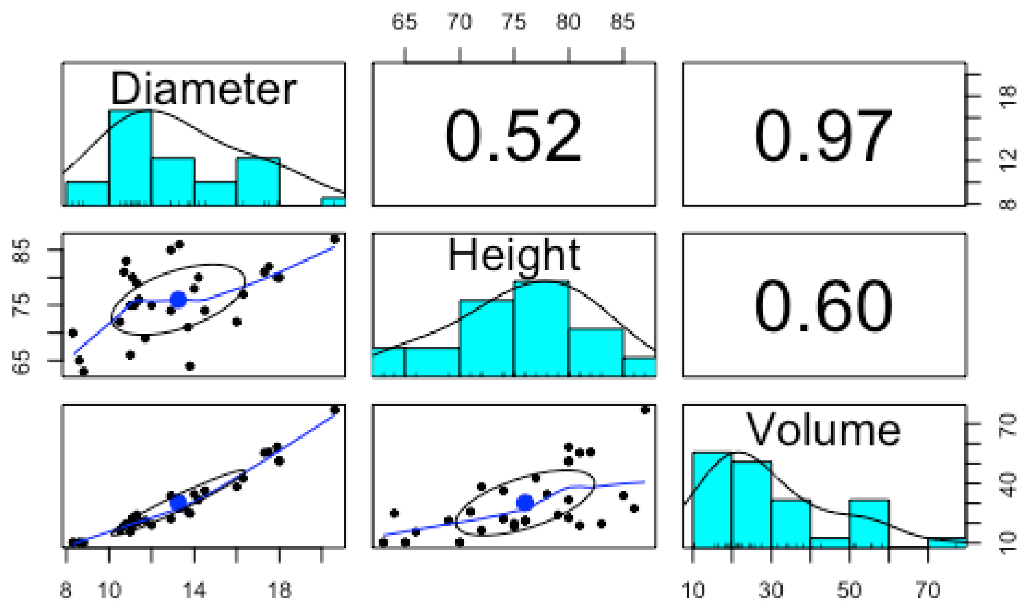
The scatterplots confirm the strong positive linear relationship between Volume and Diameter as we can see the points following a clear trend on the plot. The points in the plot between height and volume also seem to follow a weaker positive trend as they appear more scattered across the plot area.

This is similar in the scatterplot between diameter and height where the points are also quite scattered but still follow a weak positive trend.

You are welcome to use the above code to create scatterplots between each pair of variables, and calculate the correlation coefficients separately, but if you prefer something cleaner, you can also use the `pairs.panels()` function from the `psych` package which produces a neat array of plots and coefficients.

CODE

```
# not using library(psych) as I am only using the function once

# instead do the following:
psych::pairs.panels(trees)
```



We can see the function produces scatterplots and correlation coefficients between each variable, as well as histograms for each individual variable. The histograms can supplement our EDA where we are looking at the distribution for each variable.

We have identified some potential collinearity occurring between height and diameter, but we will run the models today. Next week we will cover how to manage collinearity in MLR.

> **i Checkpoint**
>
> You should now be able to describe the distribution of each variable using summary statistics, histograms and boxplots, and identify any potential outliers. You should also be able to use correlation coefficients and scatterplots to identify the strength and direction of relationships between variables, and to spot potential collinearity between predictor variables.

Exercise 3: Fit models and check assumptions

Our correlation analyses identified that Diameter is more strongly correlated with Volume, so we will use Diameter as our predictor variable in the simple linear regression.

Fitting multiple and simple linear regression uses the same function in R, you just need to add on the additional predictors for multiple linear regression.

CODE

```
mod_slr <- lm(Volume ~ Diameter, data = trees)
```

```r
mod_mlr <- lm(Volume ~ Diameter + Height, data = trees)
```

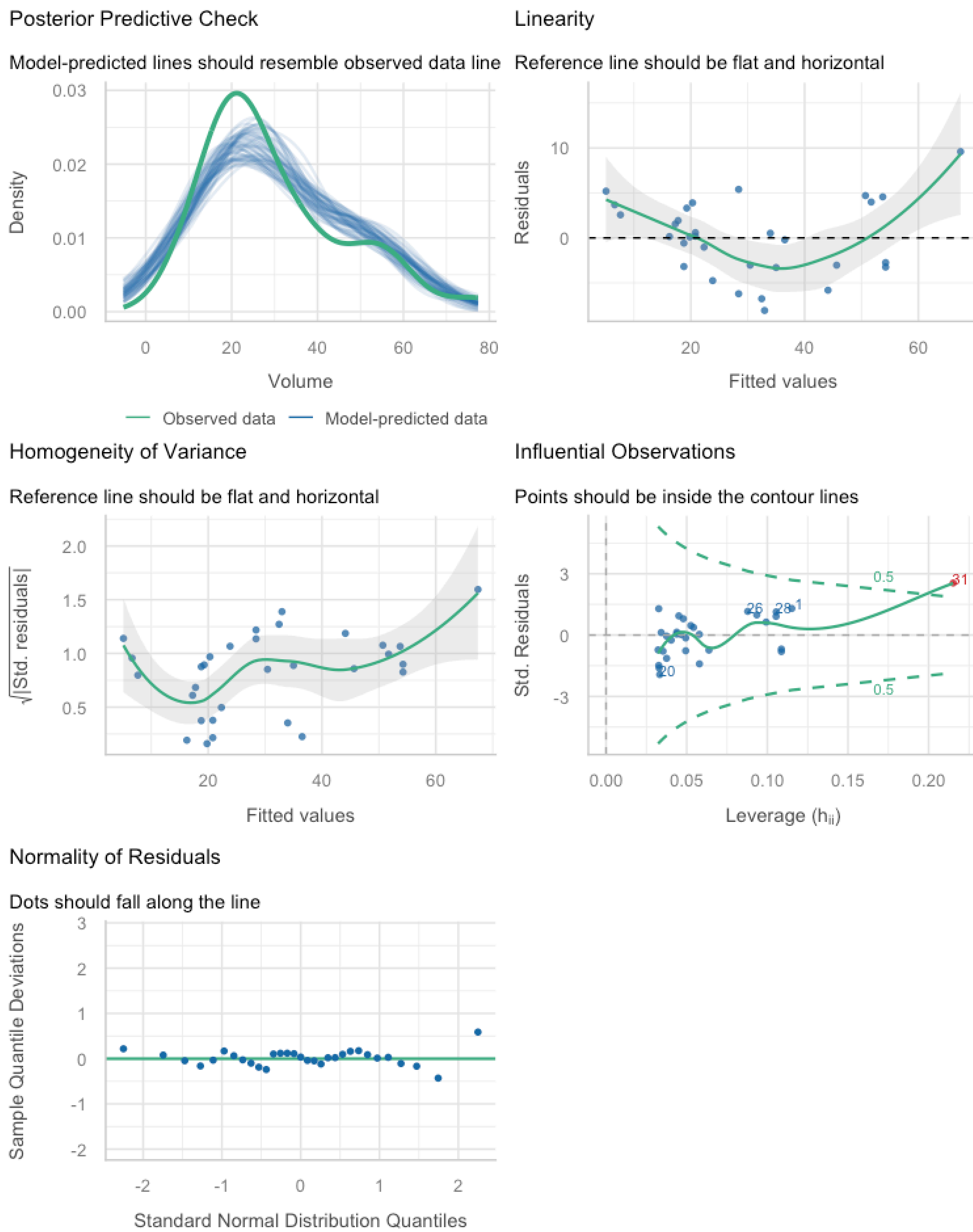
Once you have fitted the models, you then need to check the assumptions have been met.

You can check model assumptions using the `plot()` function, but the `check_model()` function from the `performance` package produces a more comprehensive set of diagnostic plots to check assumptions.

Part A — Simple linear regression

CODE

```r
library(performance)

check_model(mod_slr)
```

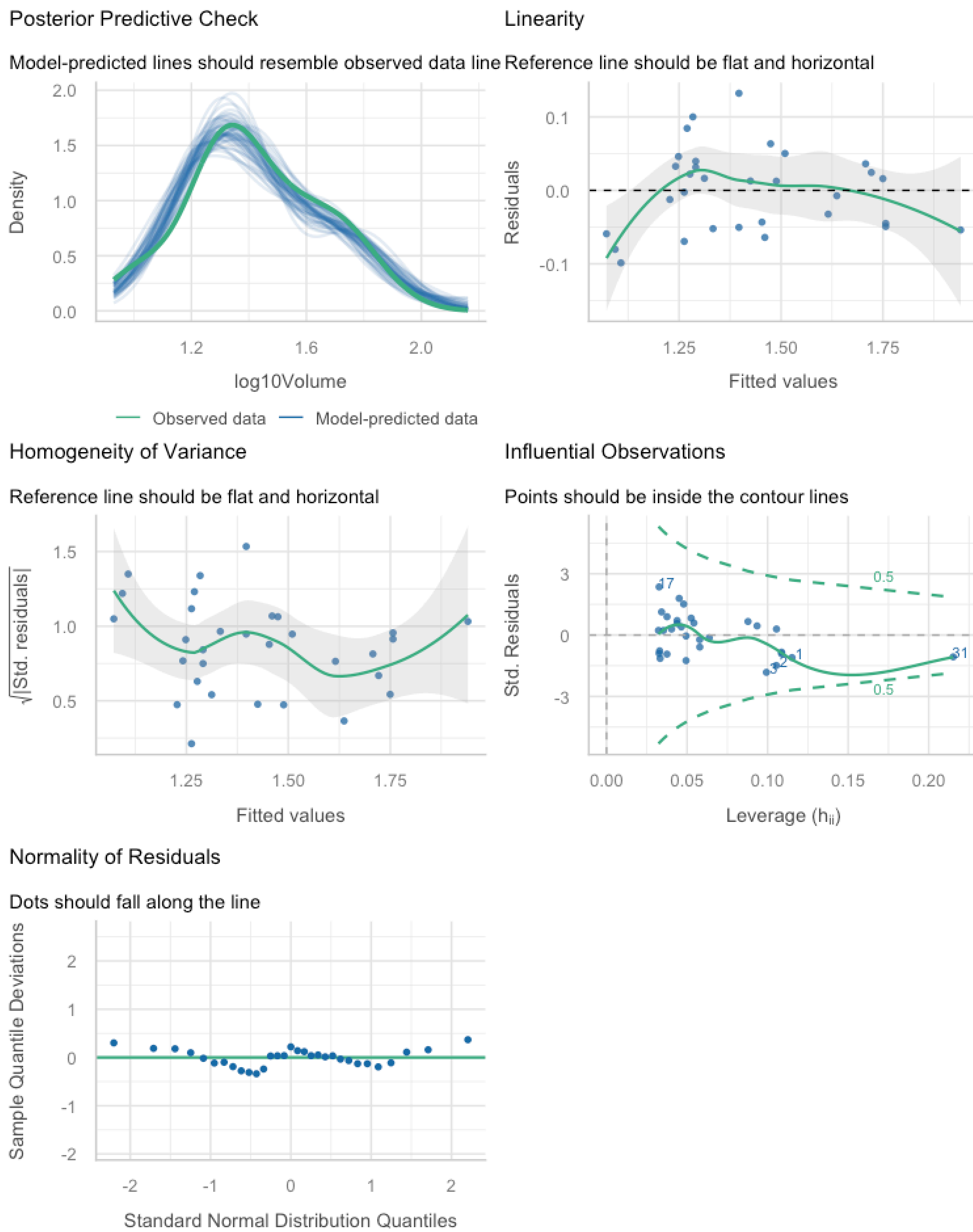


Our assumptions for a simple linear regression are L.I.N.E. - Linearity, Independence, Normality, Equal variance. We also need to check if there are any observations that are having a larger influence on the model (higher leverage).

- L: The Linearity (residuals vs fitted) plot shows the residuals are not randomly scattered around the horizontal line at 0. This tells us that there is a non-linear relationship between diameter and volume.
- I: The Independence assumption is met as the data is collected from different trees, so the observations are independent of each other.
- N: The Normality (QQ) plot shows that the residuals mostly follow the straight line, which indicates that the residuals are normally distributed. It looks like the assumption of normality has been met.
- E: The residuals vs fitted plot also shows the reference line is not flat and the points are not evenly scattered. This tells us that the variance of the residuals is not constant across all levels of diameter, which violates the assumption of equal variance (homoscedasticity).
- Influential observations: The residuals vs leverage plot shows that there is an outlier in the data which may have an influence on the model. This is likely due to the high values of volume that we identified in our EDA.

Decision: Our assumptions are not met, and we know that the volume variable is positively skewed, so we will log transform the response variable to stabilise the variance of the regression and manage the leverage of the outliers in the variable. This should reduce the skew and help meet our assumptions.

```
CODE
# Create column containing log10Volume

trees <- trees %>%
    mutate(log10Volume = log10(Volume))

# Refit model with transformed Volume

log_mod_slr <- lm(log10Volume ~ Diameter, data = trees)

# Recheck assumptions

check_model(log_mod_slr)
```



- L: The Linearity (residuals vs fitted) plot looks better, with more points following the line. The line still does not perfectly follow the horizontal line at 0, but it is much improved from the previous plot, so we can say that the linearity assumption is now mostly met. If we wanted to, it would also be worth exploring non-linear modelling but we will not cover that today.

- I: The Independence assumption is met as the data is collected from different trees, so the observations are independent of each other. (same as before; not something we can always test for and it links back to the original experimental design).
- N: The Normality (QQ) plot shows that the residuals follow the straight line much better now, which indicates that the residuals are normally distributed. It looks like the assumption of normality has been met.
- E: The standardised residuals vs fitted plot shows the reference line is flatter, does not deviate as much as before, and the points are a bit more scattered. This tells us that the variance of the residuals has improved to a point where we can continue to our analyses. If you have something like this in a report, it is worth acknowledging that other models may be more appropriate and worth exploring.
- Influential observations: The residuals vs leverage plot is no longer showing a point outside the dashed (Cook's distance) line. This means there are no longer observations that are having a higher influence on the model and we can proceed with our model interpretation.

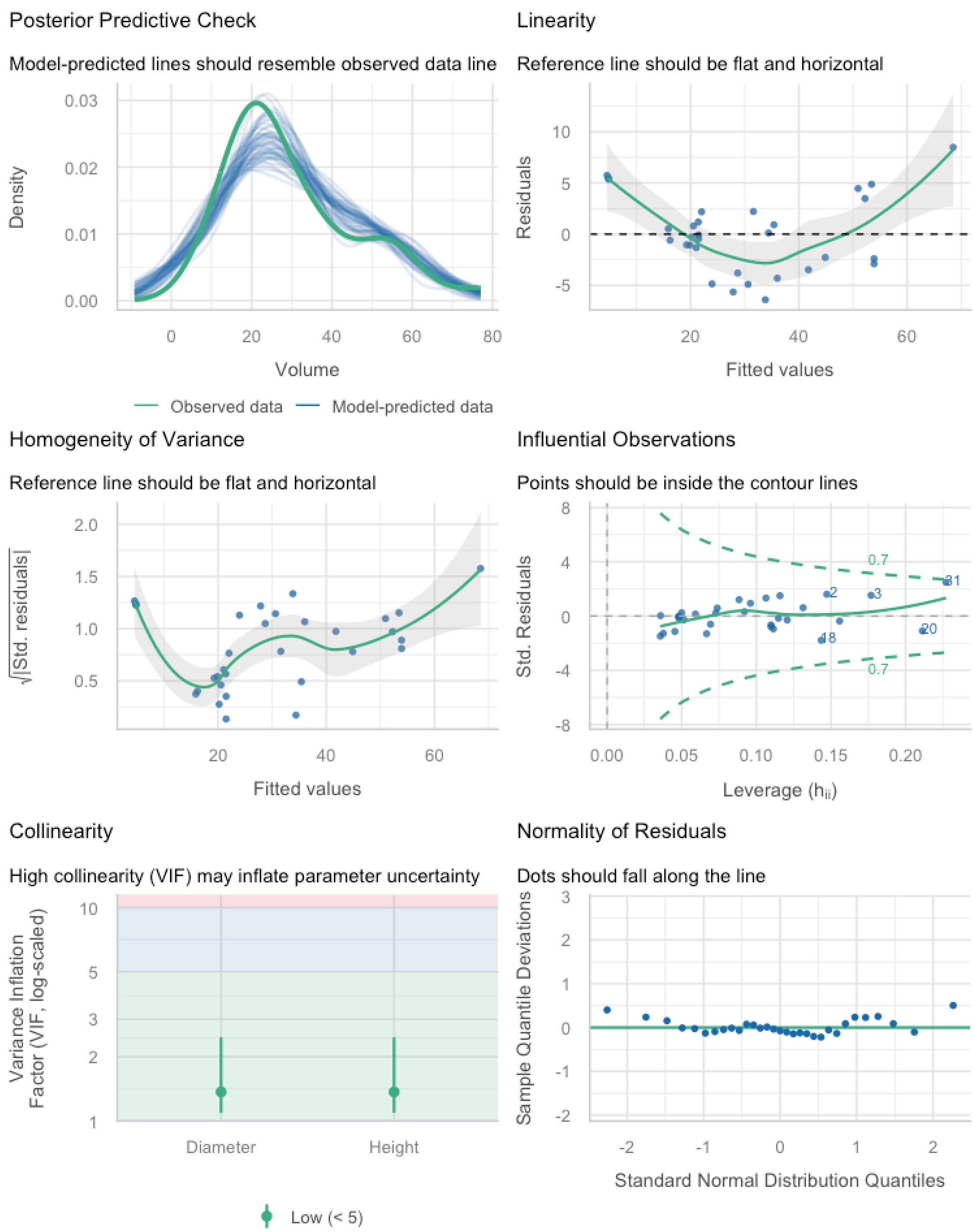
Decision: Transforming Volume has helped us meet the model assumptions.

Some of the diagnostics have told us the relationship may be non-linear, so in the real world we would also consider fitting alternative models to see if there is something better out there to represent the relationship between tree volume and diameter. This is especially important if we wanted to use the model for prediction.

Part B — Multiple linear regression

CODE

```
check_model(mod_mlr)
```



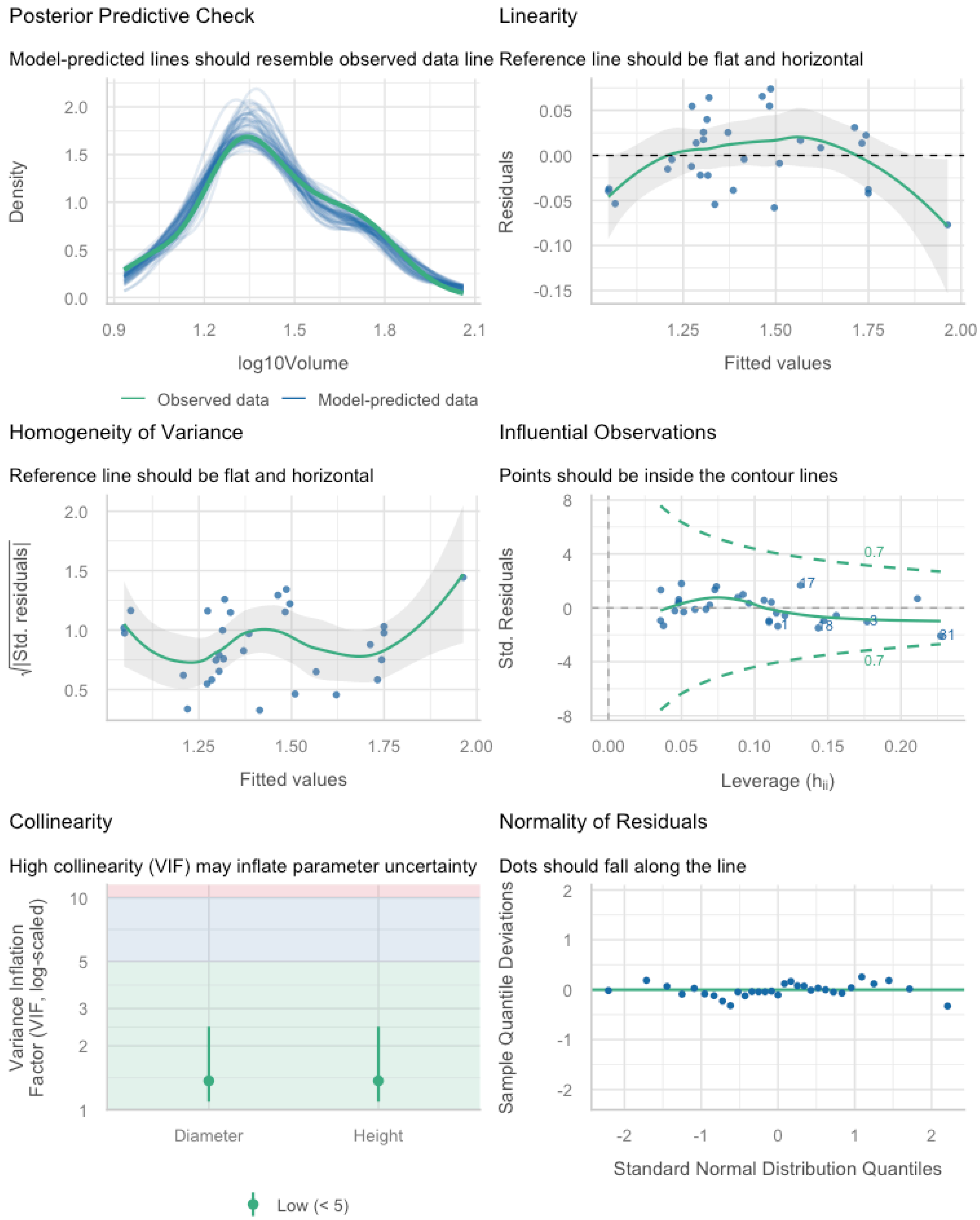
Our assumptions for a multiple linear regression are L.I.N.E. - Linearity, Independence, Normality, Equal variance AND collinearity. We also need to check if there are any observations that are having a larger influence on the model (higher leverage).

- L: The Linearity (residuals vs fitted) plot shows the residuals are not randomly scattered around the horizontal line at 0. This tells us that there is a non-linear relationship between our predictors and volume.
- I: The Independence assumption is met as the data is collected from different trees, so the observations are independent of each other. (this is same as before; not something we can always test for and it links back to the original experimental design).
- N: The Normality (QQ) plot shows that the residuals mostly follow the straight line, which indicates that the residuals are normally distributed. It looks like the assumption of normality has been met.
- E: The residuals vs fitted plot also shows the reference line is not very flat and the points are not evenly scattered. This tells us that the variance of the residuals is not constant across our predictors, which violates the assumption of equal variance (homoscedasticity).
- Influential observations: The residuals vs leverage plot shows no points lying outside the Cook's distance (dashed) line, telling us that there are no single values influencing the model.
- Collinearity: We can check for collinearity using the VIF (Variance Inflation Factor) which is provided in our `check_model()` function from the `performance` package. A VIF value above 5 (or 10, depending on who you ask) indicates a potential issue with collinearity. In our model, the VIFs for Diameter and Height are both low (< 5), which indicates that there is no issue with collinearity between these two predictor variables.

Decision: Our assumptions are not met, and we know that the volume variable is positively skewed, so like in our simple linear regression, we will use log-transformed Volume to stabilise the variance of the regression. This should reduce the skew and help meet our assumptions.

CODE

```
# Refit model with transformed Volume

log_mod_mlr <- lm(log10Volume ~ Diameter + Height, data = trees)

# Recheck assumptions

check_model(log_mod_mlr)
```



- L: The Linearity (residuals vs fitted) plot looks slightly better, with more points following the line. The line still does not perfectly follow the horizontal line at 0, but it is improved from the previous plot, so we can say that the linearity assumption is now mostly met. If we wanted to, it would also be worth exploring non-linear modelling but we will not cover that today.

- I: The Independence assumption is met as the data is collected from different trees, so the observations are independent of each other. (same as before; not something we can always test for and it links back to the original experimental design).
- N: The Normality (QQ) plot shows that the residuals follow the straight line much better now, which indicates that the residuals are normally distributed. It looks like the assumption of normality has been met.
- E: The standardised residuals vs fitted plot shows the reference line is flatter and the points are a bit more scattered. This tells us that the variance of the residuals has improved to a point where we can continue to our analyses.
- Influential observations: The residuals vs leverage plot shows the points are closer to the dashed line centred at 0 and no points are outside the dashed (Cook's distance) line, indicating that there are no individual observations impacting on the model.

Decision: Similar to the simple linear regression, transforming Volume has helped us meet the model assumptions.

Also similar to the simple linear regression, some of the diagnostics have told us the relationship may be non-linear, so in the real world we would also consider fitting alternative models to see if there is something better out there to represent the relationship between tree volume and diameter. This is especially important if we want to use the model for prediction.

> **i Checkpoint**
>
> You should now be able to test assumptions for simple linear regression and multiple linear regression.
>
> We can now move onto interpretation!

Exercise 4: Evaluate model fit

Part A — Simple linear regression conclusion

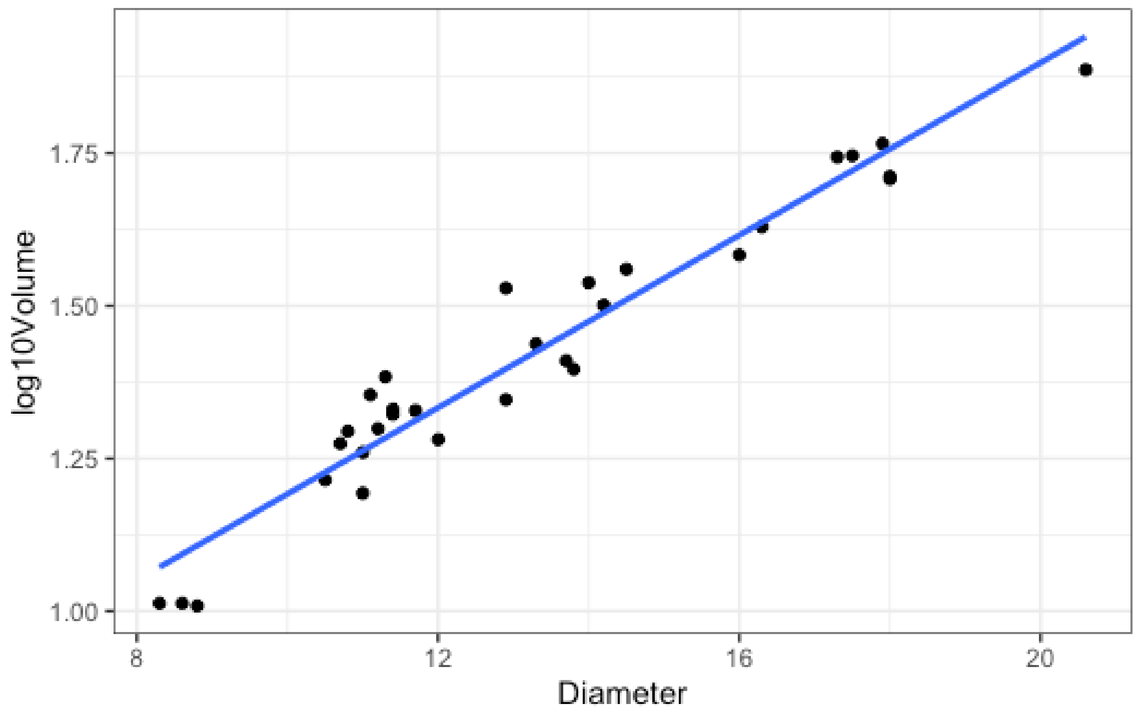
First, we can plot the model, to give some context to the reader and provide a visual representation to the model output we will be interpreting.

CODE

```
ggplot(trees, aes(x = Diameter, y = log10Volume)) +
    geom_point() +
    geom_smooth(method = "lm", se = FALSE) +
    theme_bw()
```

OUTPUT

```
`geom_smooth()` using formula = 'y ~ x'
```



Once we have created the plot, we can interpret the statistical output.

CODE

```
summary(log_mod_slr)
```

OUTPUT

```
Call:
lm(formula = log10Volume ~ Diameter, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.09867 -0.04981  0.01255  0.03444  0.13218 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.485974   0.045176   10.76 1.23e-11 ***
Diameter    0.070601   0.003321   21.26  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05708 on 29 degrees of freedom
Multiple R-squared:  0.9397,	Adjusted R-squared:  0.9376 
F-statistic:   452 on 1 and 29 DF,  p-value: < 2.2e-16
```

1. State the model equation

Model equation is $log10(Volume) = 0.486 + 0.0706 \cdot Diameter$

2. Interpretation of regression coefficients

For every 1 unit change in Diameter, we expect a 0.0706 increase in log10(Volume).

3. P-value

The P-value for the slope coefficient is < 0.001, meaning we reject our null hypothesis that the slope is equal to zero ($H_0 : \beta_1 = 0$) and conclude that Diameter is a significant predictor of log10(Volume).

This is supported by our plot as we can see the regression line is not horizontal, but has a slope to it.

4. R-squared

As we only have one predictor in the model, we can use the r^2 value rather than the adjusted r^2.

The r^2 value is 0.94, which means that 94% of the variation in log10(Volume) can be explained by Diameter. This is a very strong r^2 value, suggesting that Diameter alone is able to explain most of the variation in log10(Volume).

We can also see visual evidence of the r^2 in the plot as the points sit close to the line. If the r^2 were smaller, we would notice the points lying further away from the line or more evenly scattered about the plot area.

Part B — Multiple linear regression conclusion

It is harder to plot a multiple linear regression model because we add on a dimension with each additional predictor. While we can still plot the relationship between our response variable and each predictor variable separately, but we cannot visualise the relationship between all three variables at the same time, so we will not be providing a scatterplot here.

CODE

```r
summary(log_mod_mlr)
```

OUTPUT

```

Call:
lm(formula = log10Volume ~ Diameter + Height, data = trees)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.076991 -0.037357 -0.004312  0.025586  0.073835 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.044552   0.093510   0.476    0.637    
Diameter    0.063098   0.002861  22.057  < 2e-16 ***
Height      0.007116   0.001409   5.051 2.41e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04202 on 28 degrees of freedom
Multiple R-squared:  0.9684,	Adjusted R-squared:  0.9662 
F-statistic: 429.7 on 2 and 28 DF,  p-value: < 2.2e-16
```

1. State the model equation

Model equation is:

$$log10(Volume) = 0.0446 + 0.0631 \cdot Diameter + 0.0071 \cdot Height$$

2. Interpretation of regression coefficients

Holding all other variables constant: (very important to include this)

- For every 1 unit change in Diameter, we expect a 0.0631 increase in log10(Volume);
- For every 1 unit change in Height, we expect a 0.0071 increase in log10(Volume).

3. P-value

When assessing the p-values, we must consider the p-value associated with each partial regression coefficient, as well as the overall model.

a. Relationship between Diameter and log10(Volume):

The P-value for the partial regression coefficient is < 0.001, meaning we reject our null hypothesis that the slope is equal to zero ($H_0 : \beta_1 = 0$) and conclude that Diameter is a significant predictor of log10(Volume).

b. Relationship between Height and log10(Volume):

The P-value for the partial regression coefficient is < 0.001, meaning we reject our null hypothesis that the slope is equal to zero ($H_0 : \beta_2 = 0$) and conclude that Height is also a significant predictor of log10(Volume).

c. Overall model:

The P-value for the model is < 0.001, indicating that at least one slope is not equal to zero. This means we reject our null hypothesis that all partial slopes are equal to zero (

$$H_0 : \beta_1 = \beta_2 = 0$$

) and conclude that our model is significant.

4. R-squared

As we have more than one predictor in the model, we will need to use the adjusted r^2. We will cover the reason for this in next week's lecture.

The adjusted r^2 value is 0.97, which means that 97% of the variation in log10(Volume) can be explained by our model. This is a very strong r^2 value. Although there was some correlation between Diameter and Height in our EDA, the VIFs from our diagnostics were low, so collinearity is unlikely to be influencing our model.

> **ℹ Checkpoint**
>
> You should now be able to interpret the summary output of a fitted simple and multiple linear regression, including the model equation, the regression coefficients, the p-values and the (adjusted) r-squared. You should also be able to explain why multiple linear regression uses the adjusted r-squared rather than the r-squared, and why the interpretation of a partial regression coefficient always requires the phrase "holding all other variables constant".

Wrap-up

In this tutorial, we have walked through the processes for simple and multiple linear regression, including how to fit models, check assumptions and how to interpret the output. You will hopefully now be able to recall key differences between the two types of models, which will help direct your analysis.

You will now apply these skills in the lab. If you have any questions please reach out to your tutors.

Example exam questions

1. Describe 3 key differences between simple linear regression and multiple linear regression.
2. Describe how we would check the assumptions of a multiple linear regression. What would we do if we found that one of the assumptions was violated?
3. If you fitted a multiple linear regression and only one predictor was significant, would you still expect the model to be significant? Justify your answer (it may help to recall the hypotheses for MLR).